

SMART INDIA HACKATHON 2025



- Problem Statement ID – 25031
- Problem Statement Title- Crowdsourced Civic Issue Reporting and Resolution System
- Theme- Clean & Green Technology
- PS Category- Software
- Team ID- 16
- Team Name (Registered on portal) - Codeoholics



CivicSetu – A complete civic issue reporting system consisting of mobile app for users and web dashboard for municipal staff.

Area Indexing

Assigning rating for the areas based on resolution time and user feedbacks



Geotagged Images and Video

Automatic location linking with reported issues



Push notifications

Notifies the user at each step of resolution



Fast Response

Reports are quickly delivered to departments, ensuring a fast response



AI Powered Filtration & Routing

Filters the genuine reports and route them to respective departments



Idea Uniqueness



Feedback Loop

User can give feedback after the resolution of issue

Mobile App

Cross Platform, Light weight, Easy to use App for issue reporting

- ✓ Phone number based Login
- ✓ Aadhaar Verification
- ✓ Status Notification
- ✓ Auto - Location Detection
- ✓ Multimedia Attachments

Web Dashboard

Issue reports management portal for municipal staff

- ✓ Departmental ID Login
- ✓ Interactive Map View for issues
- ✓ Advance report filters and search
- ✓ Analytics & Metrics
- ✓ User Feedbacks Viewer

Backend

Scalable and robust backend for handling high volumes of data

- ✓ Python based backend
- ✓ Role based access
- ✓ PostgreSQL database
- ✓ Analytics engine
- ✓ Automated Routing Engine
- ✓ AI powered reports filtration
- ✓ Notifications & Communication
- ✓ Encrypted Storage
- ✓ JWT based secure API endpoints
- ✓ Location & Mapping Services

Civic Issue Categories



Water supply



Sewage



Roads



Street lighting



Waste Management



Others

Technologies

❑ User App (Cross Platform)

- **Framework** : Flutter
- **Languages** : Dart
- **Libraries** : Flutter Map, Firebase Cloud Messaging, Geolocator.

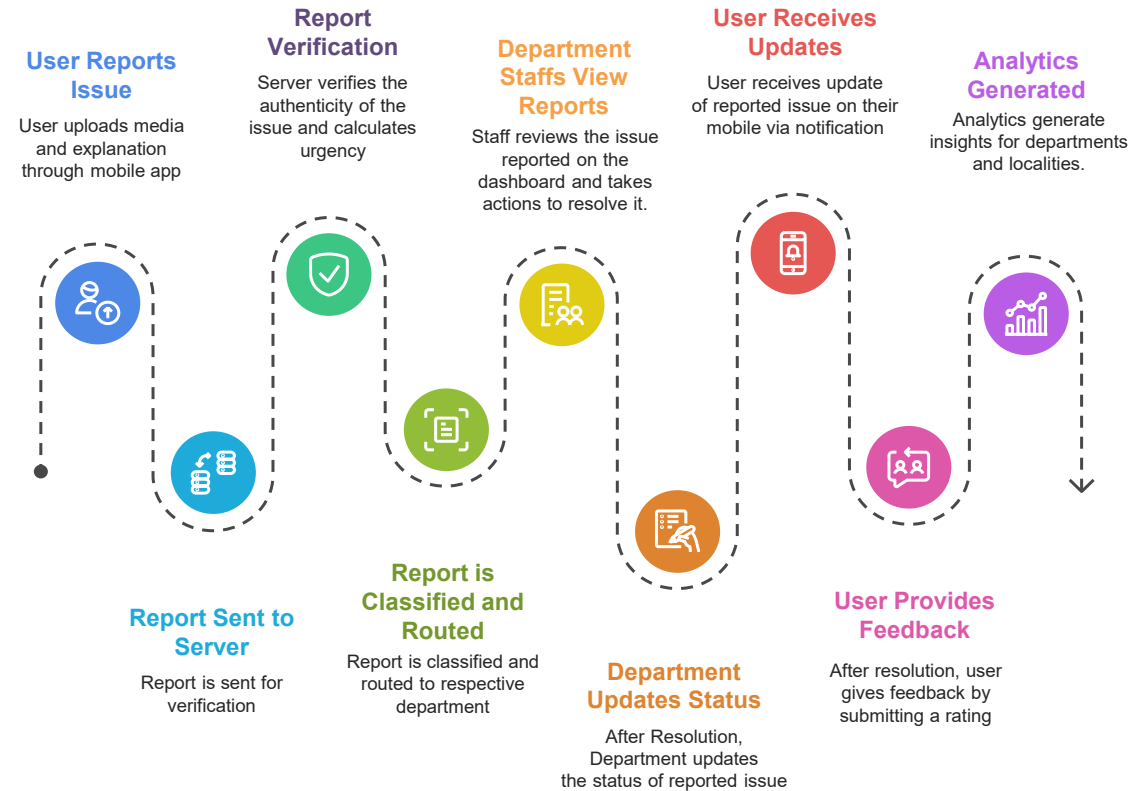
❑ Web Dashboard

- **Framework** : ReactJS, Tailwind CSS
- **Languages** : TypeScript, TSX.
- **Libraries** : Leaflet, Recharts, Axios

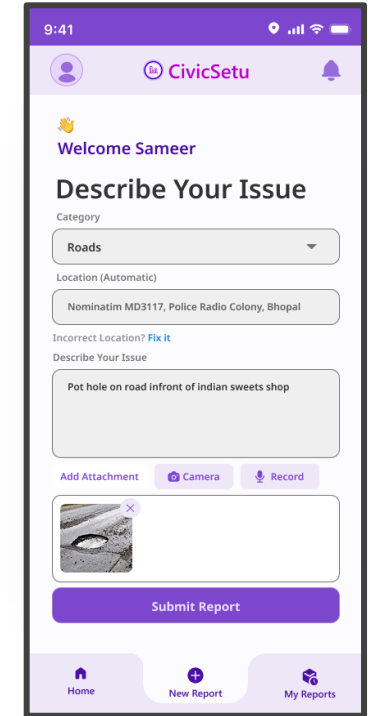
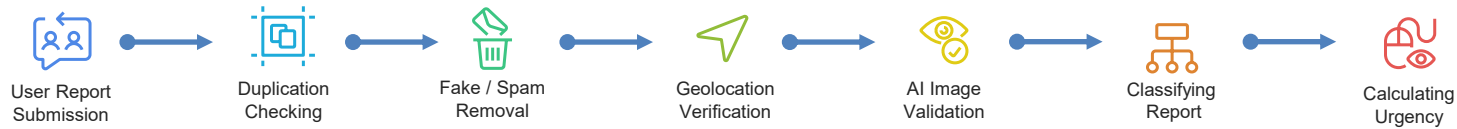
❑ Backend

- **Framework** : FastAPI
- **Languages** : Python.
- **Server** : Uvicorn
- **Libraries** : OpenCV, scikit-learn
- **Services** : OpenStreet Map, Firebase Cloud Messaging, SMS
- **Database** : PostgreSQL + PostGIS.
- **Authentication** : PyJWT
- **Storage** : S3-compatible object storage
- **Deployment** : Docker

Solution Workflow



Report Verification and Classification Process



Mobile App Prototype



Web Dashboard Prototype

Feasibility

☐ Scalable

- Built using popular widely accepted tech-stack ensures scalability
- Deployed using Docker container
- Using S3 compatible object storage makes system fast and scalable

☐ Adaptable

- Multilingual support
- User friendly interface
- User login using phone number
- Cross platform support for app

☐ Economic

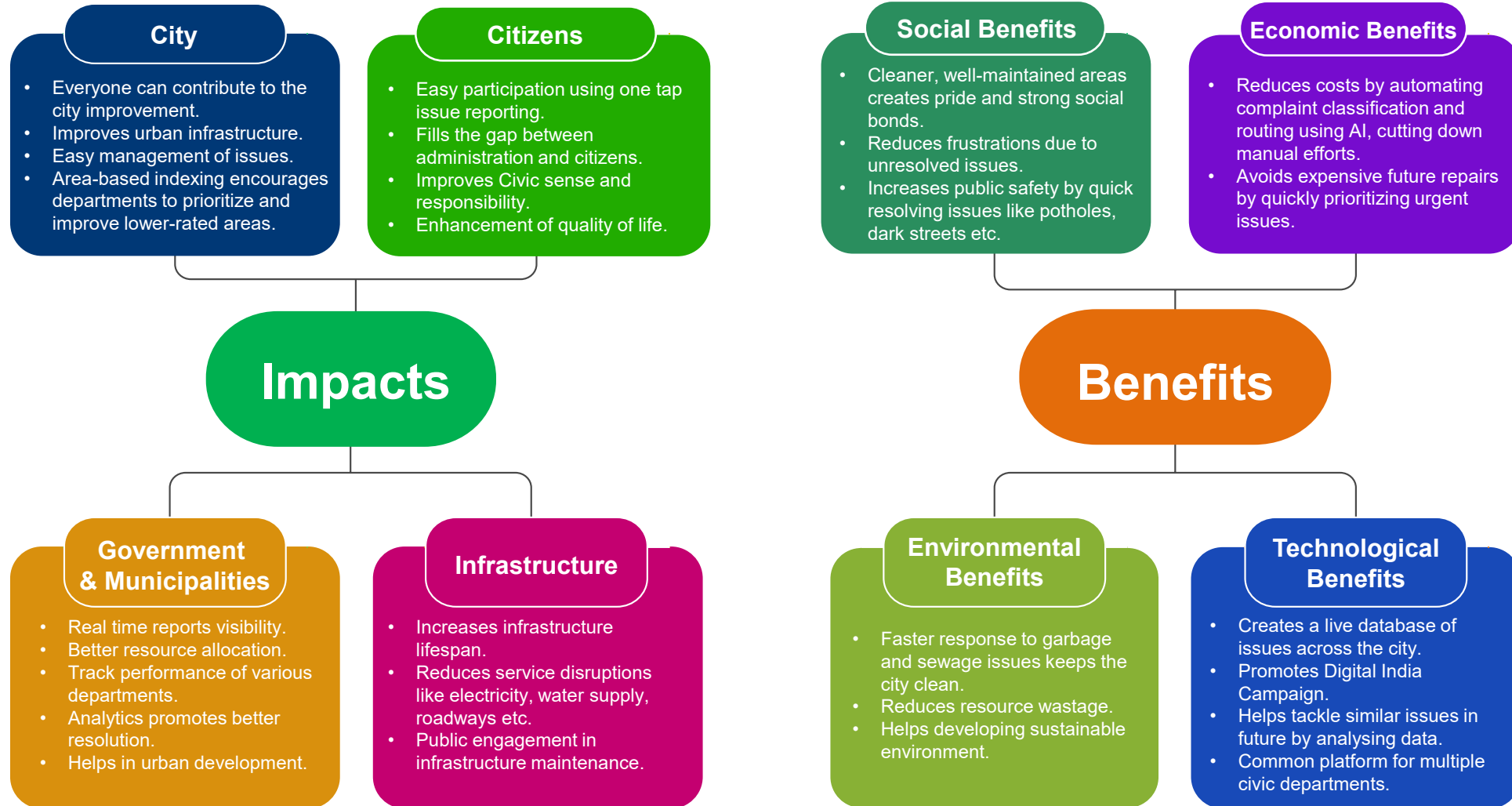
- Mostly uses open source technology
- Government support through Digital India Campaign
- Reduces manual complaint handling

Challenges and Risks



Solution Strategies

- | Challenges and Risks | Solution Strategies |
|---|---|
| <ul style="list-style-type: none">• User Engagement and Adaptation | <ul style="list-style-type: none">• Multilingual Support, Easy interface, Fast response, Reliable System. |
| <ul style="list-style-type: none">• Fake, Spam and Duplicate Reports | <ul style="list-style-type: none">• AI powered report verification and duplicates removal system. |
| <ul style="list-style-type: none">• Infrastructure and Scalability | <ul style="list-style-type: none">• Smooth deployment using Docker and future proof & scalable tech-stack. |
| <ul style="list-style-type: none">• Resistance from Departments and Delayed responses | <ul style="list-style-type: none">• Organising training workshops and following hierarchical system for fast response. |
| <ul style="list-style-type: none">• Data Security and Privacy | <ul style="list-style-type: none">• Full stack solution ensures no requirement of third party services hence improves data security and privacy |



References -

1. Municipality System in India

<https://en.wikipedia.org/wiki/Municipality>

2. Administrative System of Municipal Corporations in India -

<https://trp.org.in/wp-content/uploads/2017/09/ARSS-Vol.6-No.2-July-December-2017-pp.40-44.pdf>

3. Fast API Documentation -

<https://github.com/fastapi/fastapi>

4. Flutter – Multiplatform app framework -

<https://flutter.dev/>

5. Docker Container -

<https://www.docker.com/>

6. Impact of online participatory forums on neighborhood equity -

<https://www.sciencedirect.com/science/article/abs/pii/S0264275125000162>